

SECURITY POLICY AND PASSWORD COMPLEXITY CHECKER USING PYTHON

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1. Introduction

This Security Policy establishes the minimum requirements for safeguarding organizational information systems. It defines standards for password management, access control, and incident reporting to ensure confidentiality, integrity, and availability of data. All employees, contractors, and third-party users must comply with this policy.

2. Password Policy:

2.1 Password Creation

- Minimum length: 12 characters.
- Must include: uppercase, lowercase, numbers, and special characters.
- Avoid dictionary words, personal information, or reused passwords.
- Multi-factor authentication (MFA) is required wherever supported.

2.2 Password Management

- Passwords must be changed every 90 days.
- Default system passwords must be changed immediately upon deployment.
- Passwords must not be shared or written down in insecure locations.
- Use of password managers is recommended for secure storage.

2.3 Password Protection

- Accounts will be locked after 5 failed login attempts.
- Passwords must be encrypted in transit and at rest.
- Administrators must enforce password history to prevent reuse of the last 5 passwords.

3. Access Control Policy

3.1 User Access

- Access is granted based on the principle of least privilege.
- Role-based access control (RBAC) must be implemented.
- All accounts must be uniquely identifiable (no shared accounts).
- Temporary access must be time-bound and revoked after use.

3.2 Administrative Access

- Elevated privileges require documented approval.
- Administrative sessions must use secure protocols (SSH, HTTPS).
- Privileged accounts must be monitored and logged.

3.3 Physical Access

- Server rooms and sensitive areas require badge or biometric authentication.
- Visitors must be escorted and logged.
- Devices must auto-lock after **15 minutes** of inactivity.

4. Incident Reporting Policy

4.1 Reporting Procedures

- All employees must report suspected security incidents immediately to the IT Security Team.
- Incidents include: unauthorized access, malware infection, data leakage, or suspicious activity.
- Reports must be submitted via the official incident reporting system or hotline.

4.2 Response and Escalation

- The IT Security Team will classify incidents by severity (Low, Medium, High, Critical).
- Critical incidents must be escalated to senior management within **1 hour**.
- Evidence must be preserved for forensic analysis.

4.3 Documentation

- Each incident must be logged with date, time, reporter, and description.
- Post-incident reviews will identify root causes and corrective actions.
- Lessons learned will be shared with relevant teams to prevent recurrence.

5. Conclusion

This Security Policy ensures that organizational systems remain protected against unauthorized access and misuse. By enforcing strong password practices, strict access controls, and timely incident reporting, the organization strengthens its defense posture and promotes a culture of security awareness.

PASSWORD COMPLEXITY CHECKER BASED ON PYTHON

Source code:

```
def check_password(password):
    length_ok = len(password) >= 8
    has_upper = any(c.isupper() for c in password)
    has_lower = any(c.islower() for c in password)
    has_digit = any(c.isdigit() for c in password)
    has_special = any(not c.isalnum() for c in
password)

    print("Password:", password)
    print("Length >= 8:", length_ok)
    print("Contains uppercase:", has_upper)
    print("Contains lowercase:", has_lower)
    print("Contains digit:", has_digit)
    print("Contains special character:",
has_special)

    if all([length_ok, has_upper, has_lower,
has_digit, has_special]):
        print("✅ Password is strong\n")
    else:
        print("❌ Password is weak\n")

# Run tests
check_password("P@ssw0rd123")
check_password("weakpass")
```

index3.py > ...

```
1 def check_password(password):
2     length_ok = len(password) >= 8
3     has_upper = any(c.isupper() for c in password)
4     has_lower = any(c.islower() for c in password)
5     has_digit = any(c.isdigit() for c in password)
6     has_special = any(not c.isalnum() for c in password)
7
8     print("Password:", password)
9     print("Length >= 8:", length_ok)
10    print("Contains uppercase:", has_upper)
11    print("Contains lowercase:", has_lower)
12    print("Contains digit:", has_digit)
13    print("Contains special character:", has_special)
14
15    if all([length_ok, has_upper, has_lower, has_digit, has_special]):
16        print("✅ Password is strong\n")
17    else:
18        print("❌ Password is weak\n")
19
20 # Run tests
21 check_password("PAssword@123")
22 check_password("weakpass")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● admin@Admins-MacBook-Air PYTHON % /usr/bin/python3 /Users/admin/Documents/PYTHON/index3.py
Password: PAssword@123
Length >= 8: True
Contains uppercase: True
Contains lowercase: True
Contains digit: True
Contains special character: True
✅ Password is strong

Password: weakpass
Length >= 8: True
Contains uppercase: False
Contains lowercase: True
Contains digit: False
Contains special character: False
❌ Password is weak
```